Chapter 27

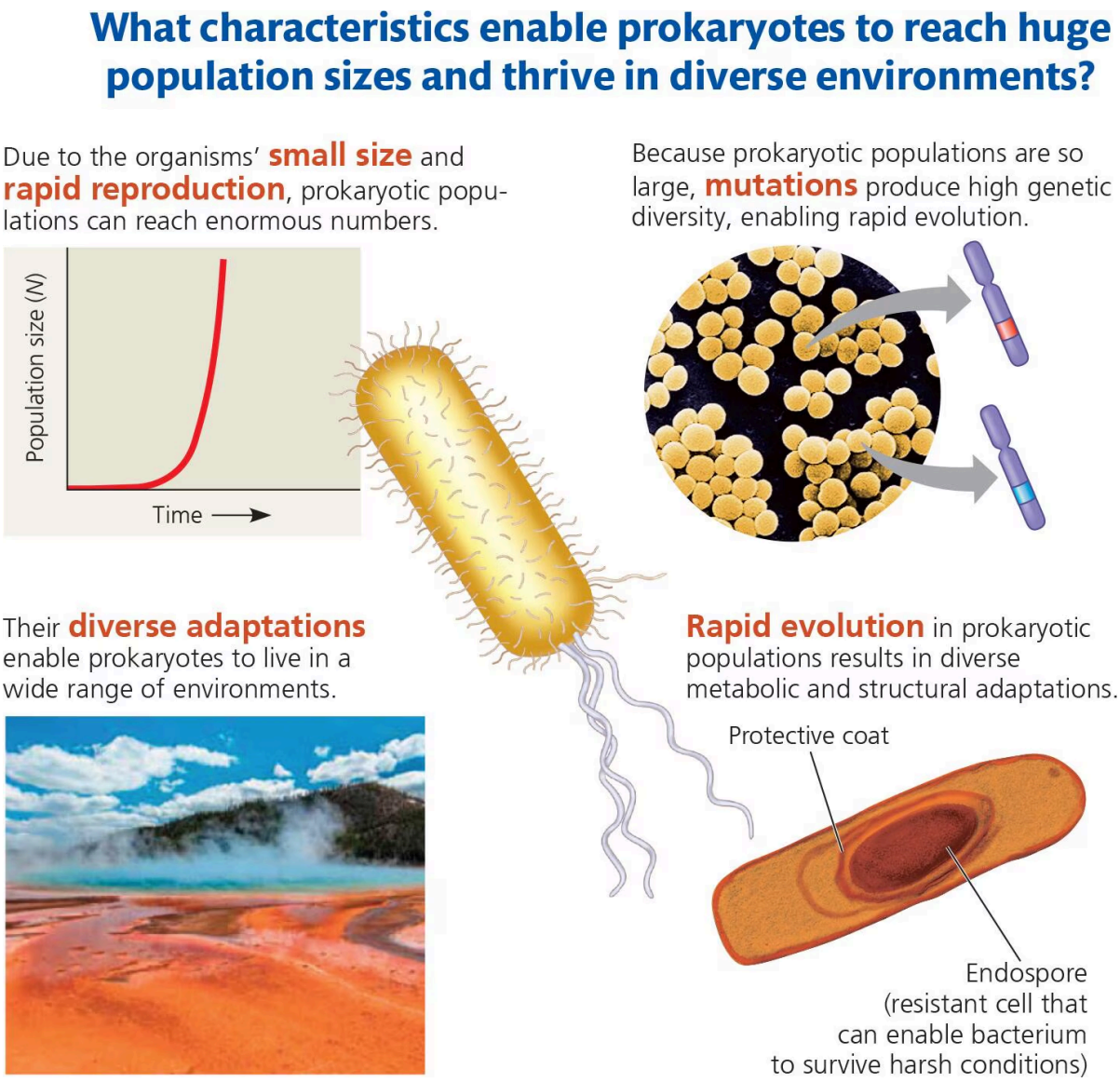
Campbell Biology, AP Edition, 12e

Chapter 27

Bacteria and Archaea

Figure 27.1

The pink color of this lake in Spain is a sign that its waters are much saltier than seawater. Its color is caused not by minerals or other nonliving sources, but by trillions of prokaryotes that thrive in salinities that kill other cells. Other prokaryotes live in environments that are far too cold or hot or acidic for most other organisms.



Due to the organisms' small size and rapid reproduction, prokaryotic populations can reach enormous numbers. Because prokaryotic populations are so large, mutations produce high genetic diversity, enabling rapid evolution. Their diverse adaptations enable prokaryotes to live in a wide range of environments. Rapid evolution in prokaryotic populations results in diverse metabolic and structural adaptations. Here, an endospore (a resistant cell that can enable a bacterium to survive harsh conditions) is surrounded by a protective coat.

Key Concepts

- 27.1**Structural and functional adaptations contribute to prokaryotic success**
- 27.2**Rapid reproduction, mutation, and genetic recombination promote genetic diversity in prokaryotes**

- 27.3**Diverse nutritional and metabolic adaptations have evolved in prokaryotes**
- 27.4**Prokaryotes have radiated into a diverse set of lineages**
- 27.5**Prokaryotes play crucial roles in the biosphere**
- 27.6**Prokaryotes have both beneficial and harmful impacts on humans**

Evidence supports the evolution of complex structures and processes in prokaryotes **(BIG IDEA 1)** that use energy to support growth, reproduction, and dynamic homeostasis **(BIG IDEA 2)**. Having evolved through rapid genetic change **(BIG IDEA 3)**, diverse forms of prokaryotes interact with other organisms and play crucial roles in the biosphere **(BIG IDEA 4)**.

Study Tip

Make a table: As you read the chapter, identify examples of challenging environments in which prokaryotes thrive. List the structural, metabolic, or other features of prokaryotes that contribute to their success in each of these environments.

Environmental challenge	Prokaryotic feature enabling success
Lack of water	Capsule or slime layer
Exposure to antibiotics	

The table is as follows:

Environmental challenge	Prokaryotic feature enabling success
Lack of water	Capsule or slime layer
Exposure to antibiotics	blank

**********Prokaryotes**, the single-celled organisms that make up domains Bacteria and Archaea, can thrive in a wide range of extreme environments. Prokaryotic species are also very well adapted to more “normal” habitats—the lands and waters in which most other species are found. Their ability to adapt to a broad range of habitats helps explain why prokaryotes are the most abundant organisms on Earth. Indeed, the number of prokaryotes in a handful of fertile soil is greater than the number of people who have ever lived. In this chapter, we’ll examine the adaptations, diversity, and enormous ecological impact of these remarkable organisms.

Get Ready for This Chapter

In this chapter, you will learn about the adaptations, diversity, and ecological impact of bacteria and archaea, the organisms referred to collectively as prokaryotes. Before beginning this chapter, you should be able to do the following:

- Differentiate between prokaryotic and eukaryotic cells (see Concept 6.2, Figure 6.5 and Figure 6.8).
- Understand how crossing over causes genetic recombination (see Concept 15.3, Figure 15.10).
- Identify sister taxa in a phylogenetic tree diagram (see Figure 26.5).

The following questions provide a quick check of your basic knowledge in these areas.

1. Which of the following features occurs in prokaryotic and in some eukaryotic cells?

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a mitochondrion
```

That's incorrect. Try again.

Incorrect. Review the following material: Only prokaryotic cells have a nucleoid. Only eukaryotic cells have a nucleus and mitochondria.

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a nucleoid
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That's incorrect. Try again.

Incorrect. Review the following material: Only prokaryotic cells have a nucleoid. Only eukaryotic cells have a nucleus and mitochondria.

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a cell wall
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That's correct.

Correct! If you had trouble with this question, review the following material: Prokaryotic cells as well as some eukaryotic cells (plants, fungi, and some protists) have a cell wall. Each of the other features listed is unique to either prokaryotic cells (nucleoid) or eukaryotic cells (nucleus and mitochondrion).

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a nucleus
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That's incorrect. Try again.

Incorrect. Review the following material: Only prokaryotic cells have a nucleoid. Only eukaryotic cells have a nucleus and mitochondria.

2. A gene that has two alleles, A and a, is located on the same chromosome as a second gene that also has two alleles, B and b. Suppose a diploid individual has alleles AB on one copy of this chromosome and alleles ab on the other copy of this chromosome. For this individual, select the pair of recombinant genotypes that could result if crossing over occurs between these two chromosomes.

AB and ab

That's incorrect. Try again.

Incorrect. Review the following material: If crossing over occurs between these loci, a portion of the chromosome containing the B allele will be exchanged with the corresponding portion of the chromosome containing the b allele.

Ab and aB

That's correct.

Correct! If you had trouble with this question, review the following material: If crossing over occurs between these loci, a portion of the chromosome containing the B allele will be exchanged with the corresponding portion of the chromosome containing the b allele. The result of this process would be to form recombinant chromosomes with the alleles Ab on one chromosome and the alleles aB on the other.

AB and aB

That's incorrect. Try again.

Incorrect. Review the following material: If crossing over occurs between these loci, a portion of the chromosome containing the B allele will be exchanged with the corresponding portion of the chromosome containing the b allele.

Ab and ab

That's incorrect. Try again.

Incorrect. Review the following material: If crossing over occurs between these loci, a portion of the chromosome containing the B allele will be exchanged with the corresponding portion of the chromosome containing the b allele.

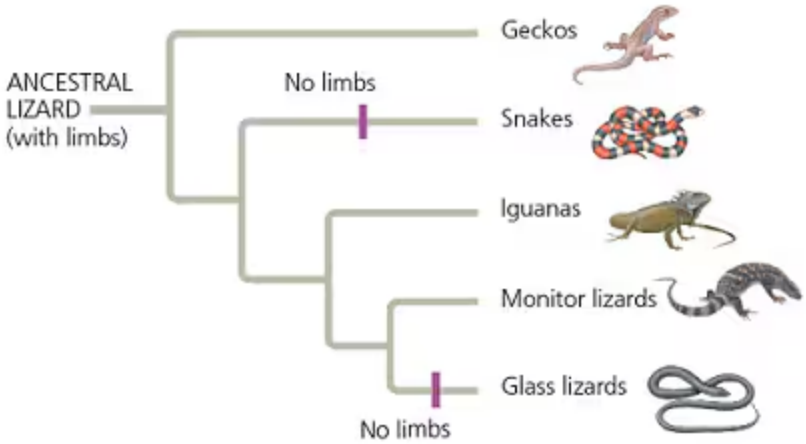

Diagram showing a series of horizontal forked branches. The leftmost end point is labeled Ancestral lizard (with limbs). Reading left to right, the first branch point forks into a horizontal line leading to Geckos and another horizontal line, which branches again. This second branch point forks into a horizontal line leading to Snakes and another horizontal line, which branches again. This third branch point forks into a horizontal

line leading to Iguanas and another horizontal line, which branches again. This fourth branch point forks into a horizontal line leading to Monitor lizards and another horizontal line, which branches again. This fifth branch point forks into a horizontal line leading to Glass lizards.

3. In the evolutionary tree shown, identify the sister taxon or taxa of iguanas.

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the group consisting of monitor lizards and glass lizards
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That's correct.

Correct! If you had trouble with this question, review the following material: Sister taxa are groups of organisms that share a common ancestor that is not shared by any other groups. In the phylogenetic tree shown, iguanas and the group consisting of monitor lizards and glass lizards share a common ancestor that is not shared by any other group.

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both snakes and the group consisting of monitor lizards and glass lizards
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That's incorrect. Try again.

Incorrect. Review the following material: Sister taxa are groups of organisms that share a common ancestor that is not shared by any other groups.

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snakes
```

That's incorrect. Try again.

Incorrect. Review the following material: Sister taxa are groups of organisms that share a common ancestor that is not shared by any other groups.

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monitor lizards
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That's incorrect. Try again.

Incorrect. Review the following material: Sister taxa are groups of organisms that share a common ancestor that is not shared by any other groups.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 27.1Describe structural and functional adaptations of prokaryotes.

- 27.2Identify sources of genetic diversity in prokaryotes.
- 27.3Give examples of nutritional and metabolic adaptations in prokaryotes.
- 27.4Identify major phylogenetic groups of prokaryotes.
- 27.5Describe ecological roles played by prokaryotes.
- 27.6Give examples of beneficial and harmful effects that prokaryotes have on humans.